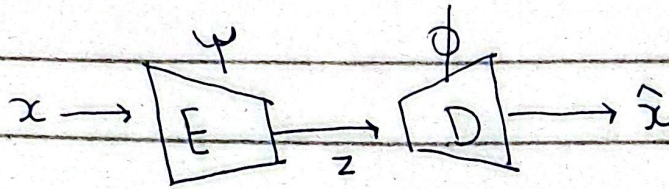


Lecture 4

Image generation:

VAE:



$$P_{\psi}(z|x) \approx q_{\psi}(z|x) \downarrow \text{gaussian}$$

$$P_{\phi}(x|z) \approx \|x - \hat{x}\|$$

Posterior

$$z \sim p(z)$$

Prior

$$N(0, I)$$

Isotropic gaussian

$$q_{\psi}(z|z)$$

Output of encoder

→ [] - mean vector

- [] - variance vector

- We can differentiate in order to back propagate because we are basically sampling something from $z \sim N(0, I)$

1. Reparameterization:

$$Z \sim \text{Stochastic Dist}$$
$$Z \sim N(0, I)$$



we basically make z deterministic

$$Z = \mu_e + \sigma_e \cdot \underline{E} \longrightarrow P(Z) = N(\mu, \sigma^2)$$

$\downarrow$
 $E \sim N(0, I)$

→ We cannot do reparameterization unless distribution is same

* same mean μ_e & σ_e

* For $Z = \mu_e + \sigma_e E + \alpha$

↓
we cannot add scalar here

2. Stop gradient → more explanation

$$y = f(\text{sg}(x))$$

function hook
on x

to be treated as constant but only during backpropagation

③ STE approx :

$$z = \text{round}(x)$$

$$\frac{\partial z}{\partial x} = 1$$

F.P. → $z = \text{round}(x)$ ← B.P.

→ more explanation

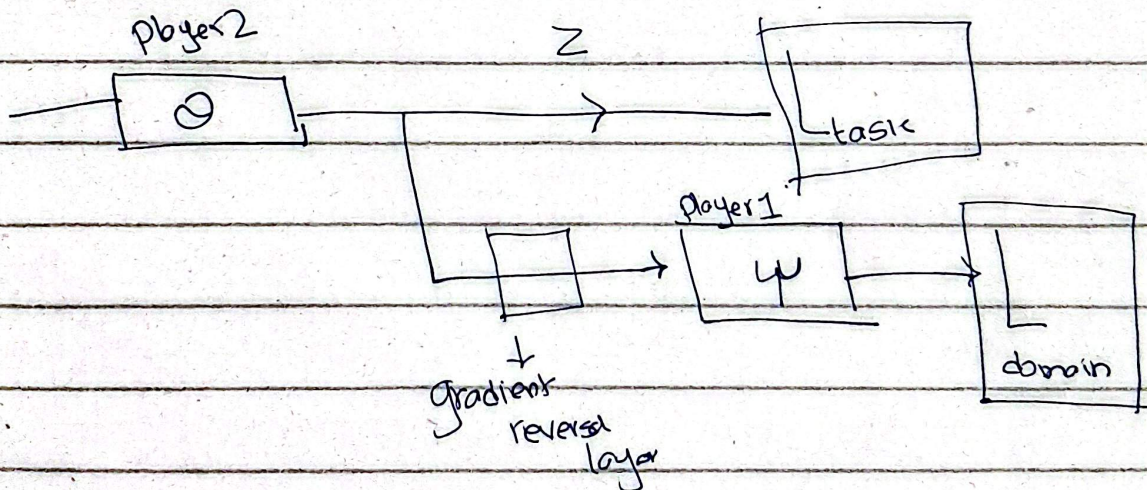
~~or~~

④ Gradient clipping:

Some gradients's evolve become high

$$g = g \times \min\left(1, \frac{C}{\|g\|}\right)$$

Hyperparameter
↓
some constant you want to tune



* adversarial learning:

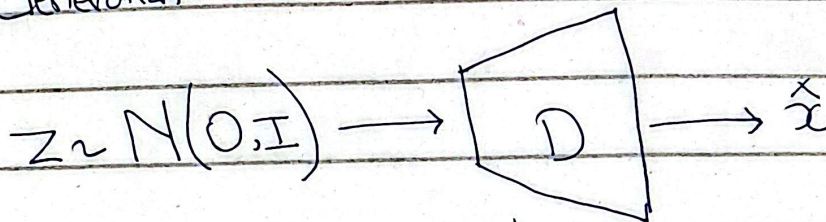
2 models — try to optimize 1 loss

- ↓
- minimise
- maximize

Gradient Reversal layer

→ GR_L will basically change the signs of gradients during back prop

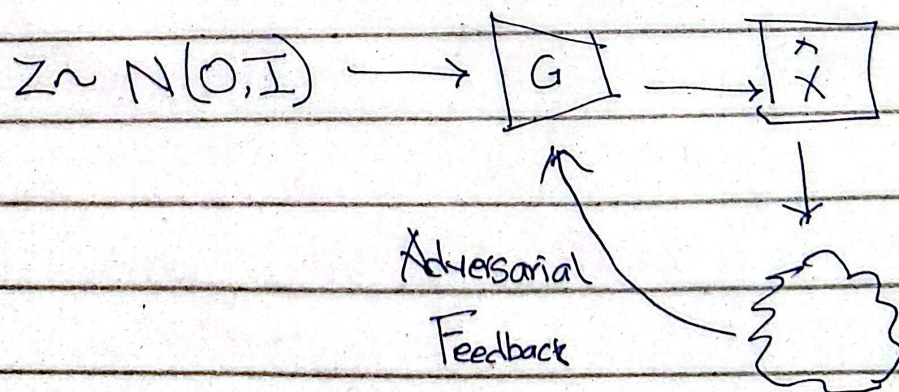
d Generation



Decode will learn during training that the mapping with help of encoder that if random sample comes from this portion → it will map to this

→ so there's no need of encoder

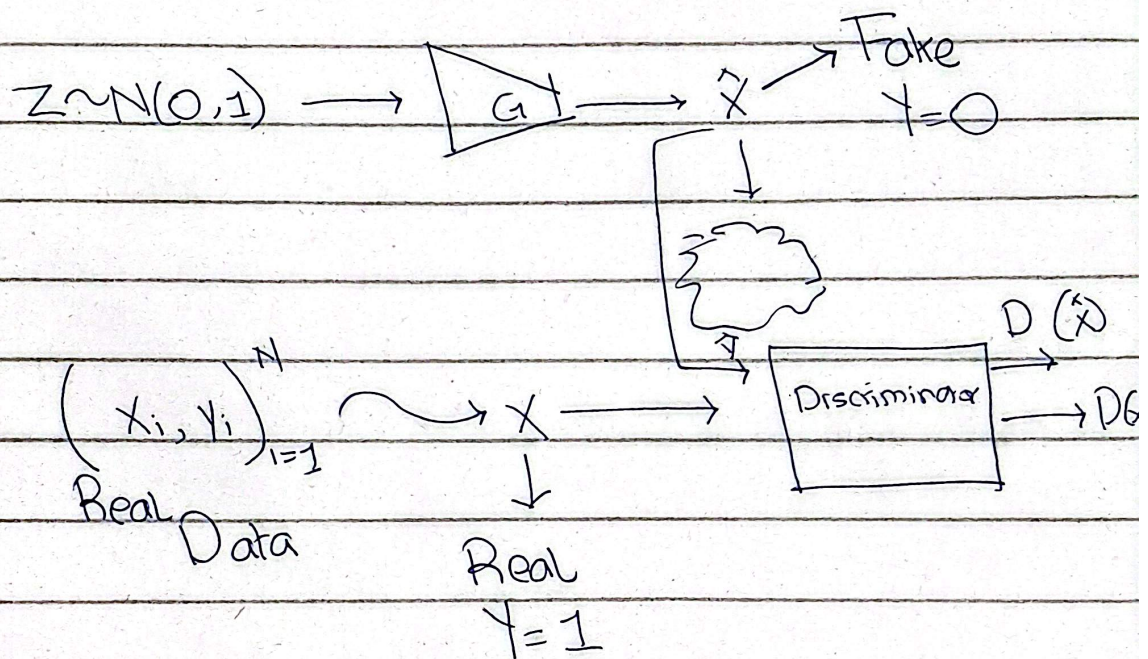
→ so, input will be pure noise



Goal: Generate Real images $\hat{x}$

Adversarial Feedback:

→ Now, we will take training examples using Real images



* $D(x)$ and $D(\hat{x})$ will be probabilities
(0,1)-Sigmoid

- Discriminator:

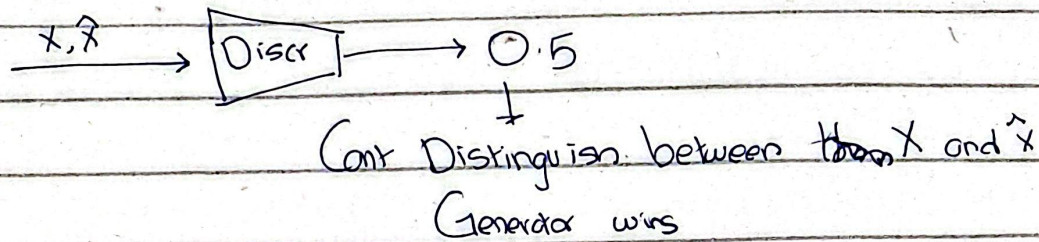
Perfect classifier between x and $\hat{x}$

↳ Binary cross entropy loss

↓
train the parameters

Discriminator: Separates between 0 and 1
Generator: Fool discriminator

Discriminator outputs probability between 0 and 1.



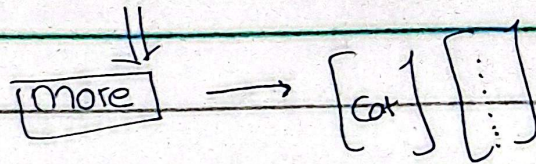
→ Suppose generator wins, and discriminator outputs 0.5. This simply means $\hat{x}$ and x are so good that the discriminator can't distinguish against which image is real → Thus $\hat{x}$ is practicing real image

$D(x)$ — Probability tells the reality of image
0.3 → 30% real image

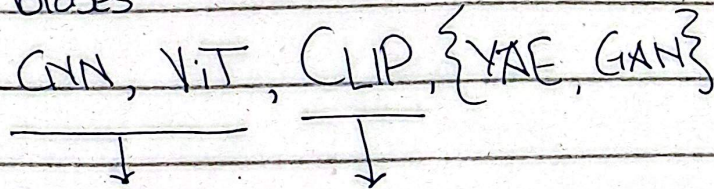
Adversarial Feedback is basically making the training loss of discriminator negative (change signs)

Mode collapse

* Diffusion model classifier Free guidance

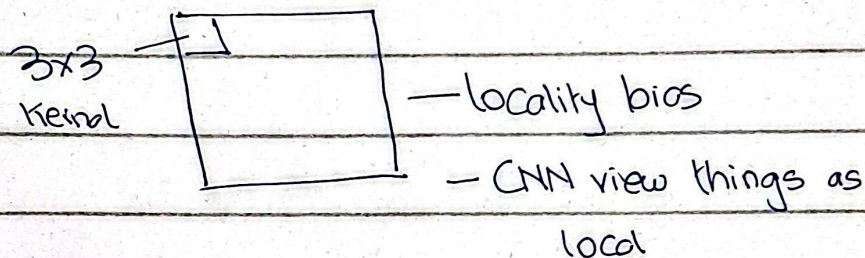


* Inductive biases



① Convolution vs. Self Attention: Architecture Bias

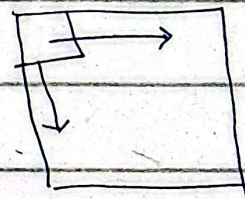
- Convolution — 1024×1024



- Self-attention

— global information

— each pixel has seen other pixels



- Which one is better

- Locality bias

- Global information

⇒ Depends on the image

↓
i) If Features are diverse across image → locality global

* Property 1 : Translation

- Translate the object either horizontal or vertical anywhere in the image

* Property 2 : Permutation

$$\begin{bmatrix} x_2 \\ x_3 \\ x_1 \end{bmatrix} = \begin{bmatrix} 0 & 1 & 0 \\ 0 & 0 & 1 \\ 1 & 0 & 0 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix}$$

$\downarrow$
 P

→ basically randomly permute the pixel of images.

⇒ Translation-invariant :

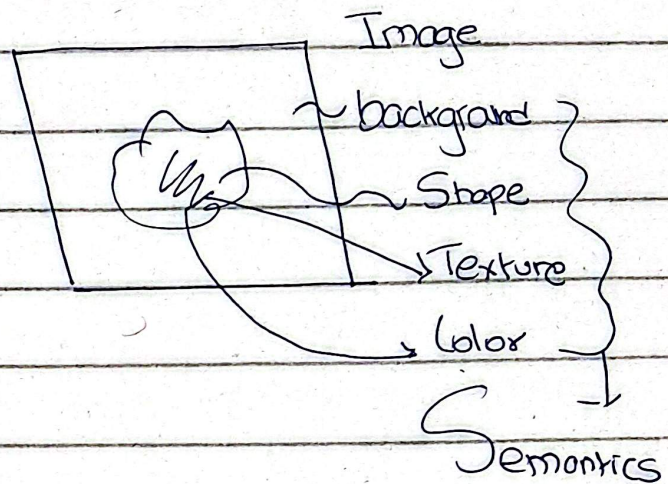
if the image is translated,
and there is no change in decision

→ Permutation-invariant

if the image is permuted, and
there is no change in decision

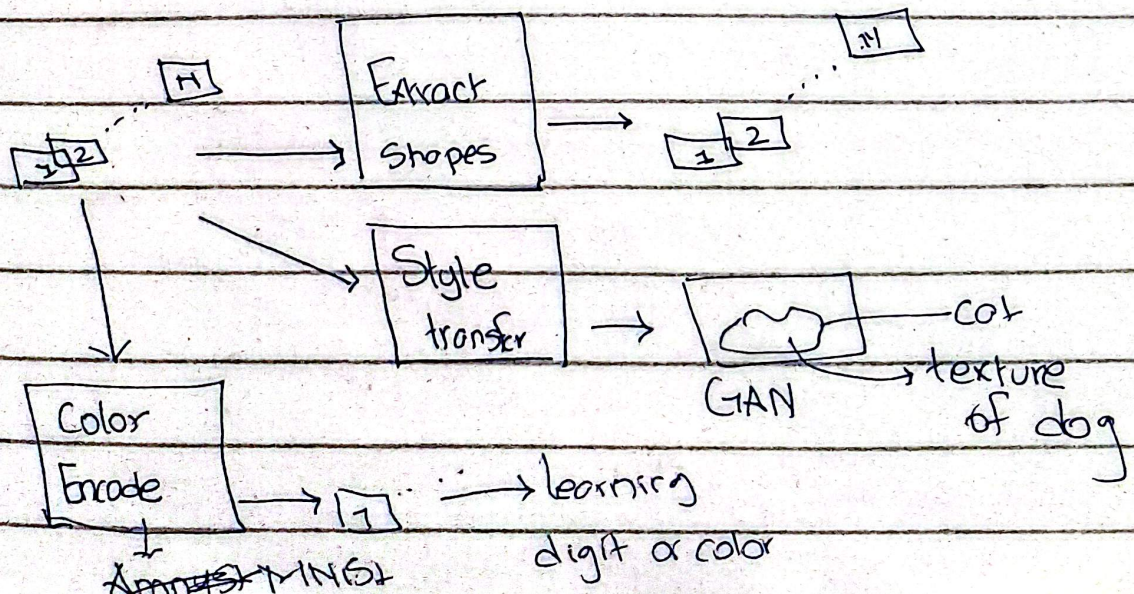
- CNN - sort of translation invariant
- pooling layers destroy it
 - ↳ randomly drop some pixels

- Semantic biases:



→ if we have k classes, and for every class, we can define $\{S_i, B_i, T_i, C_i\}_{i=1}^k$

→ When model is in inference, which of the semantics is model particularly focused on?



Results:

- CNNs — highly sensitive to texture
- Transformer — highly ~~on~~ focused on shape

Weak inductive bias — Transformers

↳ doesn't assume anything

So, they need a lot of data [Data Hungry]

- CNNs strong inductive bias — low data

↓
Distribution of your data